MEDIUM · ADVANCED MATH · ANSWER KEY

Equivalent expressions

01

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 D

D. $(x+4)^{\frac{1}{2}}$

Choice D is correct. The given expression can also be written as $(x^2+8x+16)^{\frac{1}{4}}$. The trinomial $x^2+8x+16$ can be rewritten in factored form as $(x+4)^2$. Thus, the entire expression can be rewritten as $((x+4)^2)^{\frac{1}{4}}$. Simplifying the exponents yields $(x+4)^{\frac{1}{2}}$.

Choices A, B, and C are incorrect and may result from errors made when simplifying the exponents in the expression $((x+4)^2)^{\frac{1}{4}}$.

Q2 4

The correct answer is 4. It's given that $2x^2+5x-12$ can be rewritten as $(2x-3)(x+k)$; it follows that $(2x-3)(x+k)=2x^2+5x-12$. Expanding the left-hand side of this equation yields $2x^2+2kx-3x-3k=2x^2+5x-12$. Subtracting $2x^2$ from both sides of this equation yields $2kx-3x-3k=5x-12$. Using properties of equality, $2kx-3x=5x$ and $-3k=-12$. Either equation can be solved for k. Dividing both sides of $-3k=-12$ by -3 yields $k=4$. The equation $2kx-3x=5x$ can be rewritten as $x(2k-3)=5x$. It follows that $2k-3=5$. Solving this equation for k also yields $k=4$. Therefore, the value of k is 4.

Q3

A

A. $\frac{h^{10}}{q^{14}}$

Choice A is correct. For positive values of a, $\frac{a^m}{a^n} = a^{m-n}$, where m and n are integers. Since it's given that $h > 0$ and $q > 0$, this property can be applied to rewrite the given expression as $h^{15-5}q^{7-21}$, which is equivalent to $h^{10}q^{-14}$. For positive values of a, $a^{-n} = \frac{1}{a^n}$. This property can be applied to rewrite the expression $h^{10}q^{-14}$ as $h^{10}\frac{1}{q^{14}}$, which is equivalent to $\frac{h^{10}}{q^{14}}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

9

The correct answer is 9. The given expression can be rewritten as $5x^3 - 3 + -1 - 4x^3 + 8$. By applying the distributive property, this expression can be rewritten as $5x^3 - 3 + 4x^3 + -8$, which is equivalent to $5x^3 + 4x^3 + -3 + -8$. Adding like terms in this expression yields $9x^3 - 11$. Since it's given that $5x^3 - 3 - -4x^3 + 8$ is equivalent to $bx^3 - 11$, it follows that $9x^3 - 11$ is equivalent to $bx^3 - 11$. Therefore, the coefficients of x^3 in these two expressions must be equivalent, and the value of b must be 9.

Q5

B

B. $-3k + 17r - 3$

Choice B is correct. Substituting $4k + 5r$ for a and $7k - 12r + 3$ for b in the expression $a - b$ yields $(4k + 5r) - (7k - 12r + 3)$. This expression can be rewritten as $(4k + 5r) + (-1)(7k - 12r + 3)$. By applying the distributive property, this expression can be rewritten as $4k + 5r - 7k + 12r - 3$, which is equivalent to $(4k - 7k) + (5r + 12r) - 3$, or $-3k + 17r - 3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

B

B. $7x^3 + 10$

Choice B is correct. The given expression is equivalent to $8x^3 + 8 - x^3 - -2$, or $8x^3 + 8 - x^3 + 2$. Combining like terms in this expression yields $7x^3 + 10$.

Choice A is incorrect. This expression is equivalent to $8x^3 + 8 - 2$, not $8x^3 + 8 - x^3 - 2$.

Choice C is incorrect. This expression is equivalent to $8x^3 + 8 - -2$, not $8x^3 + 8 - x^3 - 2$.

Choice D is incorrect. This expression is equivalent to $8x^3 + 8 - x^3 + 2$, not $8x^3 + 8 - x^3 - 2$.

Q7

C

C. $\frac{1}{9}$

Choice C is correct. Multiplying the given functions f and g yields $f x \cdot g x = x^2 + bx 9x^2 - 27x$. Applying the distributive property to the right-hand side of this equation yields $f x \cdot g x = x^2 9x^2 - 27x + bx 9x^2 - 27x$. Applying the distributive property once again to the right-hand side of this equation yields $f x \cdot g x = x^2 9x^2 - x^2 27x + bx 9x^2 - bx 27x$, which is equivalent to $f x \cdot g x = 9x^4 - 27x^3 + 9bx^3 - 27bx^2$. Factoring out x^3 from the second and third terms yields $f x \cdot g x = 9x^4 + -27 + 9b x^3 - 27bx^2$. Since the left-hand sides of $f x \cdot g x = 9x^4 + -27 + 9b x^3 - 27bx^2$ and $f x \cdot g x = 9x^4 - 26x^3 - 3x^2$ are equal, it follows that $-27 + 9b x^3 = -26x^3$, or $-27 + 9b = -26$, and $-27bx^2 = -3x^2$, or $-27b = -3$. Adding 27 to each side of $-27 + 9b = -26$ yields $9b = 1$. Dividing each side of this equation by 9 yields $b = \frac{1}{9}$. Similarly, dividing each side of $-27b = -3$ by -27 yields $b = \frac{1}{9}$. Therefore, the value of b is $\frac{1}{9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

C

C. $6x^2y^2x^6 + 2$

Choice C is correct. Since each term of the given expression has a common factor of $6x^2y^2$, it may be rewritten as $6x^2y^2(x^6) + 6x^2y^2(2)$, or $6x^2y^2(x^6 + 2)$.

Choice A is incorrect. This expression is equivalent to $12x^8y^2$, not $6x^8y^2 + 12x^2y^2$.

Choice B is incorrect. This expression is equivalent to $6x^6y^2$, not $6x^8y^2 + 12x^2y^2$.

Choice D is incorrect. This expression is equivalent to $6x^6y^2 + 12x^2y^2$, not $6x^8y^2 + 12x^2y^2$.

Q9 D

D. $8xx^9 - x^8 + 11$

Choice D is correct. Since $8x$ is a common factor of each term in the given expression, the expression can be rewritten as $8xx^9 - x^8 + 11$.

Choice A is incorrect. This expression is equivalent to $7x^{11} - 7x^{10} + 87x^2$.

Choice B is incorrect. This expression is equivalent to $8^{10}x - 8^9x + 88x$.

Choice C is incorrect. This expression is equivalent to $8x^{11} - 8x^{10} + 88x^2$.

Q10 B

B. $1 - p^7$

Choice B is correct. Multiplying $(1 - p)$ by each term of the polynomial within the second pair of parentheses gives $(1 - p)1 = 1 - p$; $(1 - p)p = p - p^2$; $(1 - p)p^2 = p^2 - p^3$; $(1 - p)p^3 = p^3 - p^4$; $(1 - p)p^4 = p^4 - p^5$; $(1 - p)p^5 = p^5 - p^6$; and $(1 - p)p^6 = p^6 - p^7$. Adding these seven expressions together and combining like terms gives $1 + (p - p) + (p^2 - p^2) + (p^3 - p^3) + (p^4 - p^4) + (p^5 - p^5) + (p^6 - p^6) - p^7$, which can be simplified to $1 - p^7$.

Choices A, C, and D are incorrect and may result from incorrectly identifying the highest power of p in the expressions or incorrectly combining like terms.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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